

Ray theory

Chapter 7

The previous chapter explained SH wave propagation in a two-layer medium. Although it gives us insight into the seismic wave propagation of a realistic Earth, the real Earth's structure is more complex. To address seismic wave propagation for a more realistic Earth model, the next is the multi-layer approximation of the Earth's structure.

A typical dominant frequency of teleseismic events recorded by seismometers is on the order of 1 Hz. In this case, the spatial scale of the Earth's structure is approximated to be sufficiently longer than the wavelength of the seismic wave ^{note 1)}. Figure 7.1 shows observed seismograms when the 2015 earthquake off the western coast of the Ogasawara Islands (the focal depth is deeper than 660 km) on the global scale. This figure exhibits many phases, which are classified in terms of the ray path and wave type. To interpret the seismic wave propagations, an approximation of geometrical optics is feasible. Exactly speaking, although this approximation is valid only for high-frequency limits, this approach is feasible for seismic exploration of Earth's structure in many cases practically.

Figure 7.2 shows the arrival times of seismic phases visually picked from the onset times of observed seismograms, plotted against the epicentral distance. Black represents the P-wave, and red represents the S-wave. Numerous phases (classified according to the path taken and the combination of wave types) can be seen. Since the entire waveforms are too complicated to model directly, it is easier to grasp the dominant features through simplified information: arrival times. This figure also shows that the Earth's internal structure can be approximated as one-dimensional because the measured travel time is a simple function of the epicentral distance. In addition, for P waves (at an angular distance of 100 degrees and travel time of about 13 minutes), the travel times are highly scattered, which can be interpreted as a large velocity heterogeneity at the core-mantle boundary.

A theory of wave propagation based on geometrical optics is known as Ray theory. This theory has played an important role in seismology for a long time. In particular, the ray theory for a stratified medium (multi-layer medium) is a basic theory for seismic wave propagations. This chapter gives the outline of ray theory in a stratified medium for interpreting seismic wave propagations in a realistic Earth.

^{note 1)} A wavenumber spectrum of Earth's heterogeneities shows that the long wavelength components are dominant (referred to as a "red spectrum").

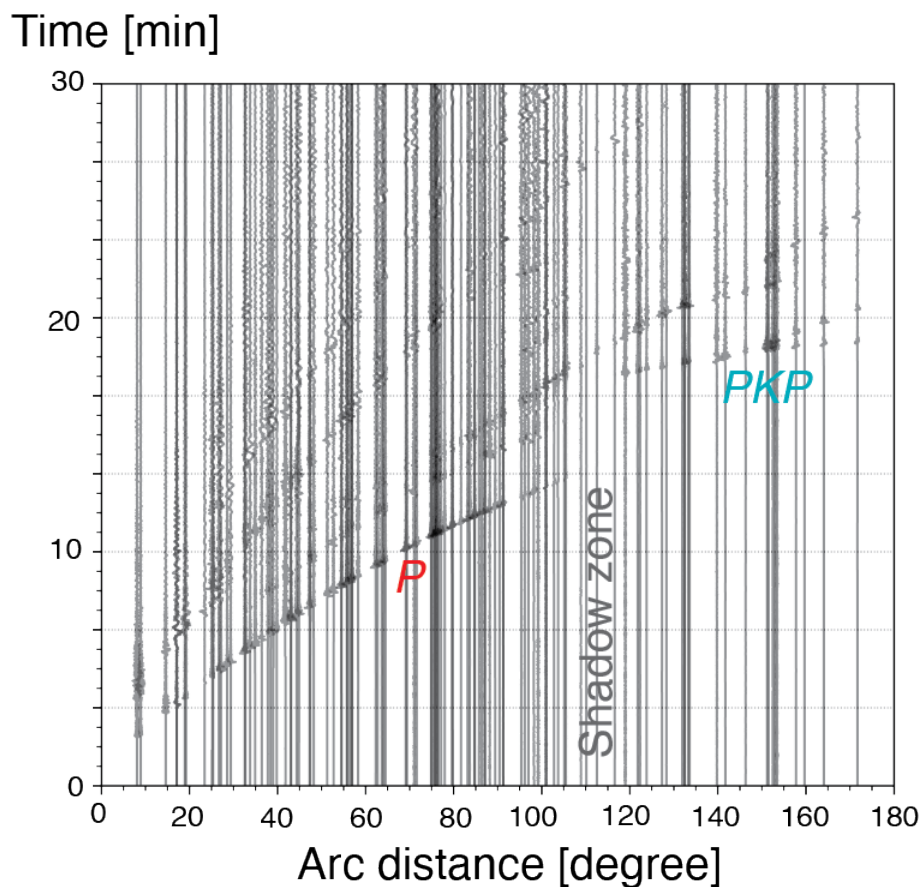


Figure 7.1: Observed records of the 2015 earthquake off the western coast of the Ogasawara Islands. The P-wave and S-wave are clear, but the waveforms are not so simple. The figure shows the shadow zone of the P-wave by the outer core, where the P-wave does not reach the surface (around the arc distance of 110 degrees).

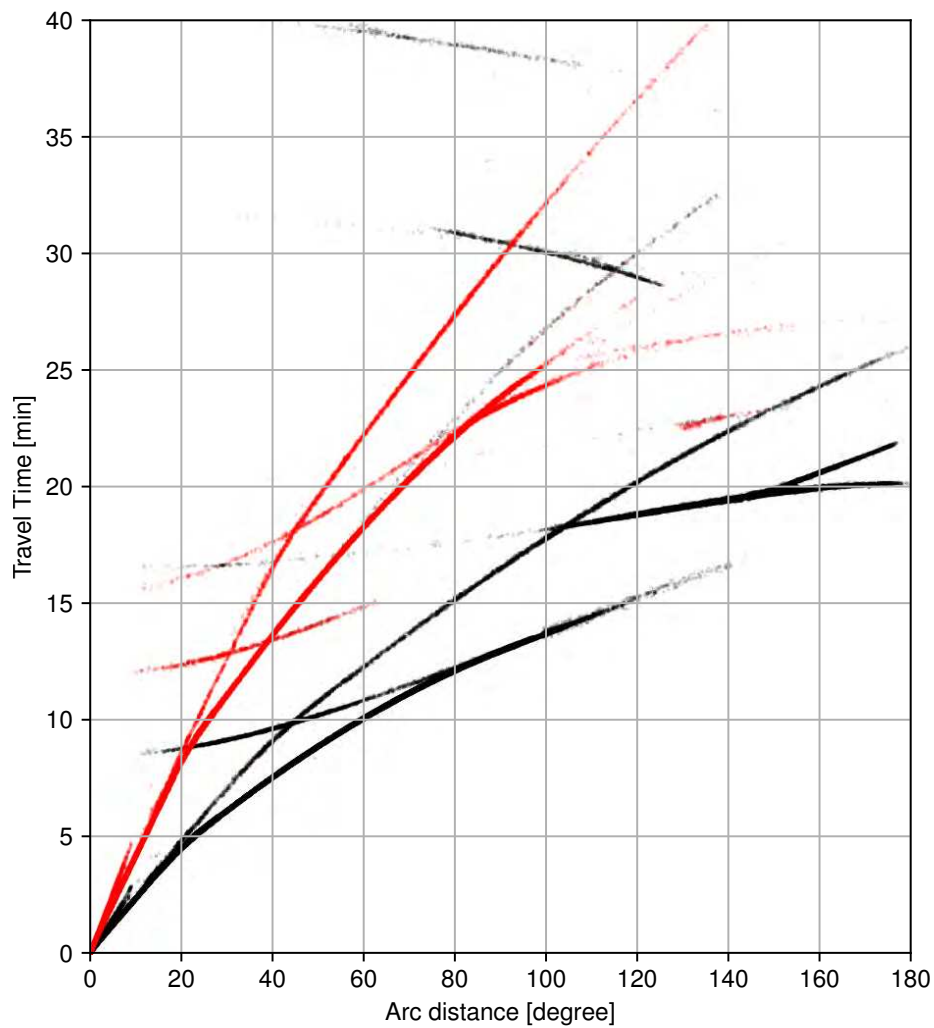


Figure 7.2: Travel-time plots of various seismic phases. Black represents P-waves, and red represents S-waves. Events shallower than 20 km that occurred in 2018 were selected. Data are from International Seismological Centre (2021), On-line Bulletin, <https://doi.org/10.31905/D808B830>.

§7.1 High frequency approximation

First, let us consider P-wave potential. With an assumption that typical spatial scales of physical properties (density and elastic constants) are enough longer than the wavelength of the P wave, the wave equation of P-wave can be simplified as

$$\ddot{\phi} = \alpha^2 \nabla^2 \phi. \quad (7.1)$$

Here we assume that a wave packet f propagates with the shape $\phi(\mathbf{x}, t) = A(\mathbf{x})f(t - T(\mathbf{x}))$. Because the potential is represented by an isosurface (wavefront), the spatial derivative is given by the gradient as $\nabla f = -\nabla T \dot{f}$. Insertion of the relation into the wave equation yields the following equation:

$$\frac{1}{\alpha^2} \ddot{\phi} = \nabla^2 A f(t - T) - 2\nabla A \cdot \nabla T \dot{f}(t - T) + A \ddot{f}(t - T)(\nabla T)^2 - A \dot{f}(t - T) \nabla^2 T. \quad (7.2)$$

Fourier transform of both sides with respect to time is written as

$$-\frac{\omega^2}{\alpha^2} A F(\omega) = \nabla^2 A F(\omega) - 2i\omega \nabla A \cdot \nabla T F(\omega) - A \omega^2 F(\omega)(\nabla T)^2 - i\omega A F(\omega) \nabla^2 T. \quad (7.3)$$

Assuming the angular frequency ω is sufficiently large, the real part of the equation yields the Eikonal equation:

$$|\nabla T|^2 = \frac{1}{\alpha^2}. \quad (7.4)$$

Here we define slowness vector $\mathbf{p}$ by ∇T .

On the other hand, the imaginary part in the high-frequency limit leads to the transport equation.

$$2\nabla A \cdot \nabla T + A \nabla^2 T = 0. \quad (7.5)$$

This equation is called a transport equation. This equation can be rewritten as $\nabla \cdot (A^2 \mathbf{p}) = 0$, which represents the conservation of energy. By solving the Eikonal equation, we get travel time T . Then we can estimate A from the transport equation with the estimated T .

§7.2 Ray tracing: Hamiltonian formalism

Based on ray theory, integration of the Eikonal equation gives travel time T . Ray tracing is one of the solving methods. Ray can be defined by successions of slowness vectors as already introduced. The wavefront expands with a speed of the seismic velocity in a direction of slowness vector. Below, the physical meaning of ray tracing is explained: ray tracing can be interpreted as tracing the motion of a particle beneath a potential. This physical system can be described more simply by using a generating parameter σ instead of time, as explained later. Although the discussion is based on analytical mechanics, the formulation is highly helpful for physical interpretation.

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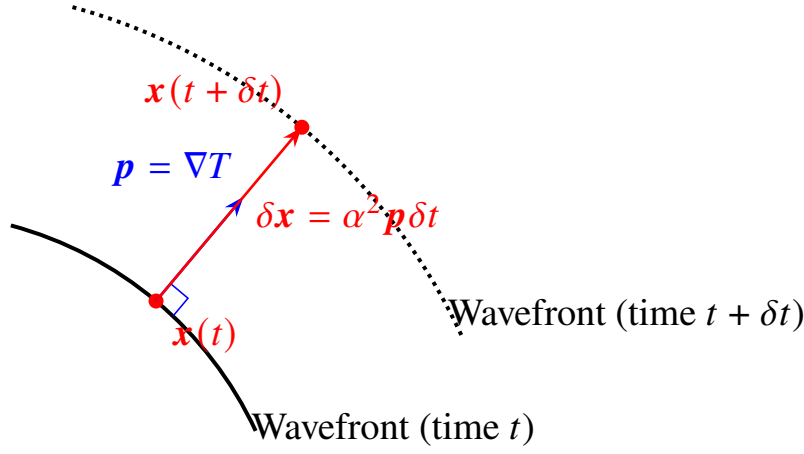


Figure 7.3: Schematic diagram of ray tracing.

7.2.1 Ray tracing

Let us consider a wavefront $T(\mathbf{x}) = t$ at time t (Figure 7.3). We trace the path of a ray from a point $\mathbf{x}(t)$ on this wavefront to the wavefront at a short time δt later, $T(\mathbf{x}) = t + \delta t$. In an isotropic medium, the ray travels perpendicular to the wavefront, along the slowness vector $\mathbf{p}$. Therefore, we have

$$\mathbf{x}(t + \delta t) = \mathbf{x}(t) + \alpha(\mathbf{x}(t))^2 \mathbf{p}(\mathbf{x}(t)) \delta t. \quad (7.6)$$

The slowness vector advances slightly to time $t + \delta t$ and changes its propagation direction. Since Snell's law holds along the ray, the change over an infinitesimal time can be written as

$$\mathbf{p}(t + \delta t) = \mathbf{p}(t) - \frac{\nabla \alpha}{\alpha} \delta t. \quad (7.7)$$

Summarizing these, we obtain a set of ordinary differential equations with respect to t :

$$\frac{d\mathbf{x}}{dt} = \alpha^2 \mathbf{p} \quad (7.8)$$

$$\frac{d\mathbf{p}}{dt} = -\frac{\nabla \alpha}{\alpha}. \quad (7.9)$$

If the slowness vector $\mathbf{p}$ is regarded as a momentum, this can be seen as describing a particle moving under a potential. However, a problem arises here. As seen from the equations, the coefficient of $\mathbf{p}$ for $\dot{\mathbf{x}}$ (which corresponds to mass) is α^{-2} , meaning it depends on the location. When viewed as a dynamical system, the mass changes with position (and thus with t). Therefore, we want to introduce a parameter equivalent to time such that the mass becomes a constant (time-invariant). We introduce the generating parameter σ as such a parameter.

7.2.2 What is a generating parameter?

This dynamical system can be described more naturally by using a generating parameter σ instead of time. We define a parameter σ (the generating parameter), which determines the particle's position corresponding to time, as a function of t . If we can define σ such that

$$\frac{d\mathbf{x}}{d\sigma} = \mathbf{p}, \quad (7.10)$$

the mass can be considered constant. That is, we consider

$$\frac{d\sigma}{dt} = \alpha^2 \quad (7.11)$$

^{note 2)} By introducing the generating parameter σ , ray tracing can be associated with the problem of tracing the motion of a particle moving under a potential $-\frac{1}{2}\alpha^{-2}$.

Hereafter, unless otherwise stated, $\dot{\mathbf{x}}$ means $\frac{d\mathbf{x}}{d\sigma}$. Given deductively, the Lagrangian L of the system is considered as

$$L(\mathbf{x}, \dot{\mathbf{x}}) = \frac{1}{2} [\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + \alpha^{-2}(\mathbf{x})], \quad (7.12)$$

which satisfies $\mathbf{p} = \partial_{\dot{\mathbf{x}}} L$ (definition of generalized momentum)^{note 3)}.

The Eikonal equation can be interpreted as a condition of the constraint of this system. The Eikonal equation can be rewritten as $H(\mathbf{x}, \mathbf{p}) = 0$ using Hamiltonian. Hamilton equations are written by

$$\frac{d\mathbf{x}}{d\sigma} = \frac{\partial H}{\partial \mathbf{p}} = \mathbf{p} \quad (7.13)$$

$$\frac{d\mathbf{p}}{d\sigma} = -\frac{\partial H}{\partial \mathbf{x}} = \frac{1}{2} \nabla \alpha^{-2}. \quad (7.14)$$

These equations can be interpreted as

1. Let us consider a problem of tracing a particle motion under a potential $-\frac{1}{2}\alpha^{-2}$.
2. The initial condition is given by $\mathbf{p}$. Note that $\mathbf{p}$ is constrained by the equation $H(\mathbf{p}, \mathbf{x}) = 0$. (This condition is equivalent that the absolute value of $\mathbf{p}$ at the initial injection point is $1/\alpha$).
3. According to Hamilton equations, $dH = \partial_{\mathbf{x}} H d\mathbf{x} + \partial_{\mathbf{p}} H d\mathbf{p} = 0$. This means that at any point along the ray path, $H = 0$ if the initial value $H = 0$ at the injection point.
4. The particle motion can be traced by integrating the Hamilton equations. Physically, σ represents time.

^{note 2)}In principle, it is possible to construct the dynamical system with s or travel time T as the parameter determining position, but we proceed with σ in this chapter because it yields the clearest equations. Although the generating parameter is less intuitive due to its analytical mechanics abstraction, it has the advantage of making the underlying theory clearer.

^{note 3)}For details, read chapter 15 of Dahlen and Tromp [1998]⁽⁵⁾

The two first-order differential equations can be simplified as a single equation:

$$\frac{d^2 \mathbf{x}}{d\sigma^2} - \frac{1}{2} \nabla \alpha^{-2} = \mathbf{0}. \quad (7.15)$$

This equation represents the equation of motion. The corresponding Lagrangian can be defined as,

$$L = \frac{1}{2} [\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + \alpha^{-2}(\mathbf{x})], \quad (7.16)$$

which satisfy the relation of $\mathbf{p} = \partial_{\dot{\mathbf{x}}} L$ (definition of generalized momentum)^{note 4)}.

Here we consider a stratified medium. In this case, because α depends only on depth z ^{note 5)}, the x component of equation ?? leads to:

$$\frac{dp_x}{d\sigma} = 0 \quad (7.17)$$

showing that p_x becomes a conserved quantity. p_x is also known as the ray parameter.

It should be emphasized here that care must be taken when handling discontinuous interfaces in the seismic velocity structure. The fact that the seismic velocity is discontinuous means that the high-frequency approximation does not hold when crossing the discontinuity. Therefore, a large error occurs if the Hamilton formalism is used for integration across a discontinuity. Consequently, when a ray crosses a discontinuity, it is necessary to consider refraction (bending the slowness vector $\mathbf{p}$) that satisfies Snell's law at the interface.

7.2.3 †Ray tracing in anisotropic media

Although details are omitted, ray tracing in anisotropic media becomes more complicated. The difficulty arises because the direction of the ray path is not perpendicular to the wavefront. Rays travel in the direction of the energy flux, but in anisotropic media, the direction of wave propagation (perpendicular to the wavefront) does not align with the direction of the energy flux. $\frac{d\mathbf{x}}{dt}$ corresponds to the energy flux, i.e., group velocity, which is no longer orthogonal to the wavefront. For details, refer to Červený's textbook.⁽¹⁰⁾

7.2.4 For spherical Earth

^{note 4)}We often use symbol $\mathbf{p}$ for slowness vector, because the slowness vector can be interpreted as generalized momentum based on ray theory. When we use time as a variable for determining the location, the generalized momentum is not proportional to the particle velocity. On the other hand, generalized momentum using generating parameter is proportional to the particle velocity. Generating parameter gives us a more simple physical analogy.

^{note 5)}In the context of analytical mechanics, such x is called a cyclic coordinate.

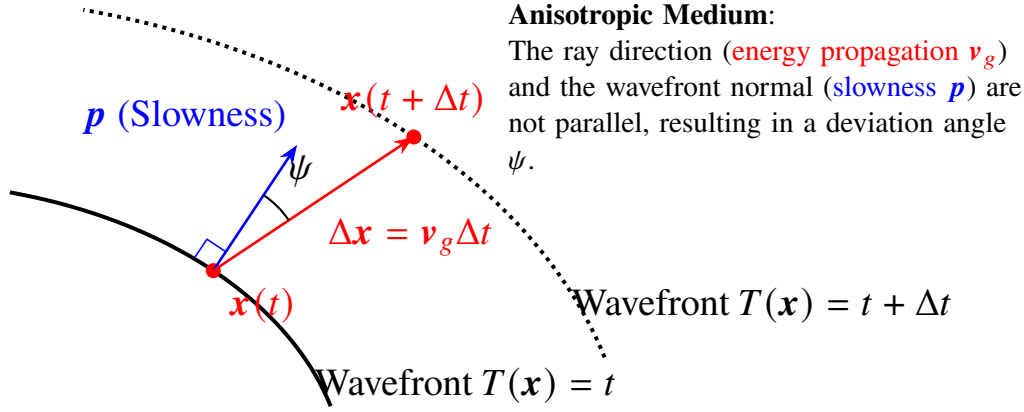


Figure 7.4: Schematic diagram of ray tracing in anisotropic media.

When we consider global seismic wave propagation, a spherical coordinate system is more convenient. With an assumption that P-wave velocity $\alpha(r)$ depend only on radius r (spherical symmetry), the Lagrangian L is written by,

$$L = \frac{1}{2}(\dot{r}^2 + (r\dot{\theta})^2 + (r \sin \theta \dot{\phi})^2 - \alpha(r)^{-2}). \quad (7.18)$$

By the appropriate choice of the spherical coordinate, we can drop ϕ . Angular distance Δ along the θ direction can describe the propagation. For a spherical structure, the slowness vector $\mathbf{p} = (p_r, p_\theta, p_\phi)$ in the spherical coordinate system is given by

$$\begin{aligned} p_r &= \frac{\partial L}{\partial \dot{r}} = \frac{dr}{d\sigma} \\ p_\theta &= \frac{\partial L}{\partial \dot{\theta}} = r^2 \frac{d\theta}{d\sigma} \\ p_\phi &= \frac{\partial L}{\partial \dot{\phi}} = (r \sin \theta)^2 \frac{d\phi}{d\sigma}. \end{aligned} \quad (7.19)$$

Similarly, the Hamiltonian H is

$$H = \frac{1}{2}(p_r^2 + r^{-2}p_\theta^2 + (r \sin \theta)^{-2}p_\phi^2 - \alpha(r)^{-2}) \quad (7.20)$$

and the Hamilton equation can be written as

$$\begin{aligned} \frac{dr}{d\sigma} &= \frac{\partial H}{\partial p_r} = p_r, \\ \frac{d\theta}{d\sigma} &= \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{r^2}, \\ \frac{d\phi}{d\sigma} &= \frac{\partial H}{\partial p_\phi} = \frac{p_\phi}{(r \sin \theta)^2}, \end{aligned} \quad (7.21)$$

$$\begin{aligned}
\frac{dp_r}{d\sigma} &= -\frac{\partial H}{\partial r} = \frac{1}{2} \frac{\partial \alpha^{-2}}{\partial r} + \frac{1}{r^3} \left[p_\theta^2 + \frac{p_\phi^2}{(\sin \theta)^2} \right], \\
\frac{dp_\theta}{d\sigma} &= -\frac{\partial H}{\partial \theta} = \frac{1}{2} \frac{\partial \alpha^{-2}}{\partial \theta} + \frac{1}{r^2} \frac{\cot \theta}{(\sin \theta)^2} p_\phi^2, \\
\frac{dp_\phi}{d\sigma} &= -\frac{\partial H}{\partial \phi} = \frac{1}{2} \frac{\partial \alpha^{-2}}{\partial \phi},
\end{aligned} \tag{7.22}$$

The result is as follows. In the case of a horizontally stratified structure, placing the emergence point at the pole does not lose generality. Also, since α does not depend on θ and ϕ , its partial derivative is 0. Assuming propagation in the direction $\phi = 0$, $p_\phi = 0$ from the definition of general momentum. Consequently, $p_\theta = 0$. Therefore, as in the Cartesian coordinate case, p_θ is conserved along the ray path, and it is also called the ray parameter.

7.2.5 Earth flattening transform

When considering ray theory, there is a simple correspondence between horizontally stratified structures ($\alpha(z)$ is a z -only function) and spherically symmetric structures. Once one problem is solved, the other can be solved by variable transformation (Earth flattening transform). Let us consider this variable transformation below.

Let us consider the problem in the θ direction for isotropic seismic wave radiation from a point. The Eikonal equation in spherical coordinates leads to the following equation.

$$\left(\frac{\partial T}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial T}{\partial \theta} \right)^2 = \frac{1}{\alpha(r)^2} \tag{7.23}$$

Using the Earth radius R_e , we can rewrite the equation as

$$\frac{1}{R_e^2} \left(\frac{\partial T}{\partial \theta} \right)^2 + \frac{r^2}{R_e^2} \left(\frac{\partial T}{\partial r} \right)^2 = \frac{r^2}{R_e^2} \frac{1}{\alpha(r)^2}. \tag{7.24}$$

We can transform the equation in spherical coordinates to an equation in Cartesian coordinates with the change of variable by $x = R_e \theta$, $r = R_e e^{-z/R_e}$, $\alpha(z) = R_e \alpha(r)/r$. The Earth flattening transform is strictly valid within ray theory. It also holds for SH wave propagation, but only approximately for SV wave. Footnotesees Aki and Richards Box 9.9 for details⁽¹⁾.

Problem 7.1

Consider a sphere with uniform velocity and find the corresponding horizontal stratification structure using the Earth flattening transform. Also, consider a wavy line in the Cartesian coordinate system and consider its correspondence with the case of a uniform sphere.

7.2.6 Fermat's principle (principle of least action)

Fermat's principle (principle of least action) states that the ray path is the path which minimizes the integration of Lagrangian among the possible paths. This leads to the following relation:

$$\delta \int_{\sigma_1}^{\sigma_2} L(\mathbf{x}, \dot{\mathbf{x}}) d\sigma = \delta \int_{\sigma_1}^{\sigma_2} [\mathbf{p} \cdot \dot{\mathbf{x}} - H(\mathbf{x}, \mathbf{p})] d\sigma. \quad (7.25)$$

This relation is equivalent to minimizing the corresponding travel time.

$$\delta \int_{x_1}^{x_2} \mathbf{p} \cdot d\mathbf{x} = \delta \int_{T_1}^{T_2} dT = 0 \quad (7.26)$$

This indicates that the ray chooses the path that minimizes the travel time. In the problem we are considering, since it physically chooses the path with the shortest travel time, the action integral turns out to be structurally equal to the total travel time, regardless of whether we choose the travel time T , the ray path length s , or the generating parameter σ as the parameter.

7.2.7 Direct method for the Eikonal equation

A method known for directly solving the Eikonal equation uses the principle of least action. This takes advantage of the fact that the ray selects the path with the shortest travel time among all possible paths. Let us consider the two-dimensional case. We cover the entire space with grid points and consider the situation where waves propagate outward from the source. At each step, for each ray path candidate that can be the shortest path, we calculate the travel time along that ray path. Then, we select the ray path with the minimum travel time. This operation is repeated until the wavefront reaches the observation station. By doing this, the travel time at the observation station can be calculated.

Assuming a constant seismic velocity simply within a grid reduces accuracy, so a higher-precision algorithm known as fast marching is known and widely used for complex (non-horizontally-stratified) structures (e.g., Rawlinson and Sambridge 2005⁽⁷⁾).

§7.3 $\tau - p$ (Radon) transform

Travel time curves against epicentral distance become complex due to the multi-valued function. There exists a transform known as $\tau - p$ (Radon) transform into the single-valued function. Here we define intersection time τ by $\tau(p) = T - p_x X$ (this type of transform is known as Legendre transforms in physics). After the transformation from a pair of T and p_x to a pair of τ and p_x , τ becomes the single-valued function even in a case with the positive jump as shown in 7.9.

Once we obtain τ against the ray parameter p_x , we can calculate the travel time T against the epicentral distance X by a relation of $d\tau/dp = -X$ ^{note 6)}.

τ is also a useful observable for array analysis of seismic data (A slant stack method of array analysis is a technique of data processing that utilizes the information from densely distributed seismometers at around X). $\tau - p$ transform is a theoretical background of the array analysis^{note 7)}

We can extend $\tau - p$ into two dimensions. To define a new variable pair that has the same information to travel time T against the dependent variable X , we must consider Legendre transform^{note 8)}.

$$\tau = T - \mathbf{p} \cdot \mathbf{X} \quad (7.28)$$

For the transform from $\mathbf{p}$ to $\mathbf{X}$,

$$\frac{d\tau}{dp_x} = -X, \quad (7.29)$$

$$\frac{d\tau}{dp_y} = -Y, \quad (7.30)$$

$$(7.31)$$

is useful^{note 9)}.

§7.4 Amplitude: geometrical spreading

^{note 6)}From the definition, the derivative of τ with respect to the ray parameter p_x is given by

$$\frac{d\tau}{dp_x} = \frac{dT}{dp_x} - X - p_x \frac{dX}{dp_x} = \frac{dT}{dp_x} - X - \frac{dT}{dX} \frac{dX}{dp_x} = -X. \quad (7.27)$$

^{note 7)}For details on practical data analysis, read a textbook by Zhou (Practical Seismic Data Analysis⁽¹²⁾).

^{note 8)}Strictly speaking, the Legendre transform requires convexity with respect to the dependent variable. Therefore we need to divide T into retrograde and prograde branches before the transform

^{note 9)}This relation can be proven as in a one-dimensional case.

As in a two-layer medium, the conservation of energy along a ray path leads to a theory for amplitudes of a ray theoretical Green's function for phases (e.g. P wave and S wave).

Here we consider seismic wave propagations in a stratified Earth model. When a seismic wave is radiated with the emergency angle of i_0 from the source, the infinitesimal solid angle $d\Omega_0 = \sin i_0 di_0 d\phi_0$ gives the conservation of energy. Here r_1 is the radius of the Earth, and ϕ is azimuth. When a radiated seismic wave with a solid angle reaches a surface point with angular distance Δ from the source,

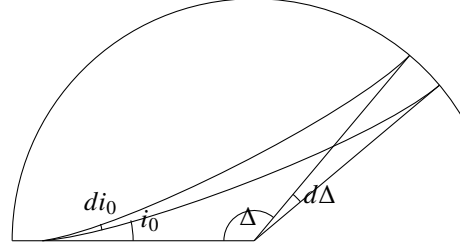


Figure 7.5:

The cross-section area on the surface S_1 is given by

$$dS_1 = r_1^2 \sin \Delta |d\Delta| d\phi_0 \cos i_1. \quad (7.32)$$

From the conservation of energy along the ray path, we obtain

$$E_0 d\Omega_0 = E_1 dS_1. \quad (7.33)$$

The energy at the surface point (the observed station) E_1 is represented by

$$E_1 = E_0 \frac{d\Omega_0}{dS_1} = E_0 \frac{\sin i_0 di_0}{r_1^2 \cos i_1 \sin \Delta |d\Delta|}. \quad (7.34)$$

Here

$$p = \frac{r_0}{\alpha_0} \sin i_0 = \frac{dT}{d\Delta} \quad (7.35)$$

$$\frac{dp}{d\Delta} = \frac{r_0}{\alpha_0} \frac{d \sin i_0}{d\Delta} = \frac{d^2 T}{d\Delta^2} \quad (7.36)$$

$$\frac{d \sin i_0}{d\Delta} = \frac{\alpha_0}{r_0} \frac{1}{\cos i_0} \frac{dp}{d\Delta}, \quad (7.37)$$

where r_0 is the distance between the source and the center of the Earth. Then we obtain geometrical spreading $\mathcal{R}$, which gives information on the amplitudes as,

$$\frac{1}{\mathcal{R}^2} = \frac{E_1}{E_0} = \frac{1}{r_1^2} \frac{\alpha_0}{r_0} \frac{\tan i_0}{\cos i_1 \sin \Delta} \left| \frac{d^2 T}{d\Delta^2} \right| \quad (7.38)$$

$$= \frac{1}{r_1^2} \frac{\alpha_0^2}{r_0^2} \frac{p}{\cos i_0 \cos i_1 \sin \Delta} \left| \frac{dp}{d\Delta} \right|. \quad (7.39)$$

$dp/d\Delta$ becomes larger with increasing ray density (see Figures in the previous section. This can be interpreted as (1) energy particles are radiated at the source toward every direction, (2) ray represents the trace of the particle (3) how many particles reach the observed station (4) by counting the energy particle, we can infer the amplitude of the phase.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

If the heterogeneities of the medium near the source are weak, we can approximate the Green's function with that of an infinite homogeneous medium. By connecting the Green's function for an infinite homogeneous medium to the ray-theoretical Green's function, we can obtain a complete ray-theoretical Green's function, including absolute amplitudes. When a station is located on the free surface, which doubles the amplitudes, the ray-theoretical Green's function of a direct P wave is given by

$$G(\mathbf{x}_r, \mathbf{x}_s, f) = \frac{1}{4\pi\mathcal{R}} \left[\frac{\hat{\boldsymbol{\eta}}_1 \hat{\boldsymbol{\eta}}_2 e^{-2\pi f i T_p}}{\sqrt{\rho_1 \rho_2 \alpha_1^3 \alpha_2}} \right], \quad (7.40)$$

where $\hat{\boldsymbol{\eta}}_1$ and $\hat{\boldsymbol{\eta}}_2$ are polarization vectors at the source and the station, respectively, and T_p is P-wave travel time.

§7.5 Caustic

When

$$\left| \frac{d^2 T}{d\Delta^2} \right| = \infty, \quad (7.41)$$

ray density diverges. This causes divergence of the amplitude in a framework of ray theory. For example, Figure 7.6 shows the concentration of ray paths at the center. This situation is analogous to lighting a fire by focusing sunlight with a magnifying glass. In this case, rays converge at a point known as a "focus". In the case of caustic, the rays focus on a line (e.g. in the case of Figure 7.6, perpendicular to the page, patterns of the ray are homogeneous).

In a region where the spatial scale of amplitude variations is comparable to the wavelength, geometrical ray theory is not appropriate. At such singular points, we need to connect the analytic solution of the original wave equation. When a ray path passes a caustic, it causes a phase jump of 90° . For example, the phase of the PP wave (reflected P wave at the surface) actually is shifted 90° , because it passes caustic one. The wave packet of the PP wave can be represented by the Hilbert transform of the direct P wave mathematically.

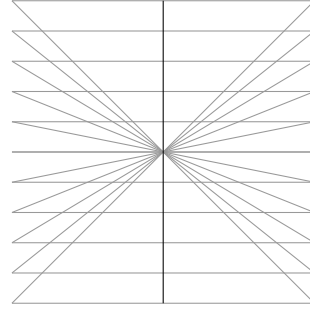


Figure 7.6: Conceptual diagram of a caustic. One can see ray paths concentrating at a point.

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \phi^2} = -k_0^2 f. \quad (7.42)$$

With the ray approximation, we can assume that typical spatial scales are sufficiently larger than the wavelength, but there exist regions where r is shorter than the wavelength in realistic situations. For simplicity, let us consider the Fourier component $f \propto e^{in\phi}$ with respect to ϕ . Then we have

$$\frac{\partial^2 f}{\partial r^2} + \frac{1}{r} \frac{\partial f}{\partial r} = \left(-k_0^2 + \frac{n^2}{r^2} \right) f. \quad (7.43)$$

This is exactly the equation satisfied by the Bessel function. The Bessel function in far-field can be approximated as

$$J_n(kr) \propto \frac{1}{\sqrt{kr}} \cos \left(kr - n\frac{\pi}{2} - \frac{\pi}{4} \right). \quad (7.44)$$

We can see that $\pi/4$ is shifted by 2 times once it passes through the origin (caustic). This is because the change in curvature works better than the derivative with respect to r when viewed in polar coordinates. Here is a rough estimate. For example, in Cartesian coordinates $\cos(\sqrt{2}/2 k_0 x) \cos(\sqrt{2}/2 k_0 y)$ satisfies the wave equation. If the wavelength corresponding to k_0 is λ_0 , the area of this positively inflated region can be written as $(\sqrt{2}\lambda_0/2)^2 = \lambda^2/2$. Considering that this is the same as the area of the circle centered at the origin, its radius is $\lambda/(\sqrt{2\pi})$. In other words, the first zero crossing point from the origin in polar coordinates is around 0.39 wavelengths away, which is more extended than $\lambda_0/4$ when considered in plane waves. This summation can be interpreted as giving the phase difference $\pi/4$ ^{note 10)}.

This phase shift can be interpreted from the conservation of energy with a natural extension of the negative cross section dS_2 .

$$E_1 dS_1 = E_2 dS_2 \quad (7.45)$$

^{note 10)} For an accurate evaluation, we need to approximate the singularity by the Airy function using the WKBJ approximation (see Arfken and Weber [2013]⁽²⁾ for details). A more physical explanation (corresponding to diffraction) is given in Landau-Lifshitz's classical field theory §59

leads to $A_2 = A_1 i$. This originated from the flip of the ray coordinates before and after a passage at a caustic. In a case of a passage at a focus, the amplitude reverses because the phase flips twice ($i^2 = -1$) according to the two directions.

An example of a caustic is the PP wave, whose phase is shifted relative to the direct P wave. This typical example is intuitively confusing, so it is explained below. Since the ground reflection complicates the behavior, let us consider a mirror image of the hypocenter with respect to the surface (Figure 7.7). If we fold back the ray path of the PP wave radiated downward from the hypocenter at the surface, we can see that it intersects before the observed station. On the other hand, the pP wave radiated from the hypocenter does not intersect. Therefore, the waveforms of the pP and direct-directed waves have the same shape, but the PP wave has a distorted shape to the direct P wave (they are related by the Hilbert transform to each other).

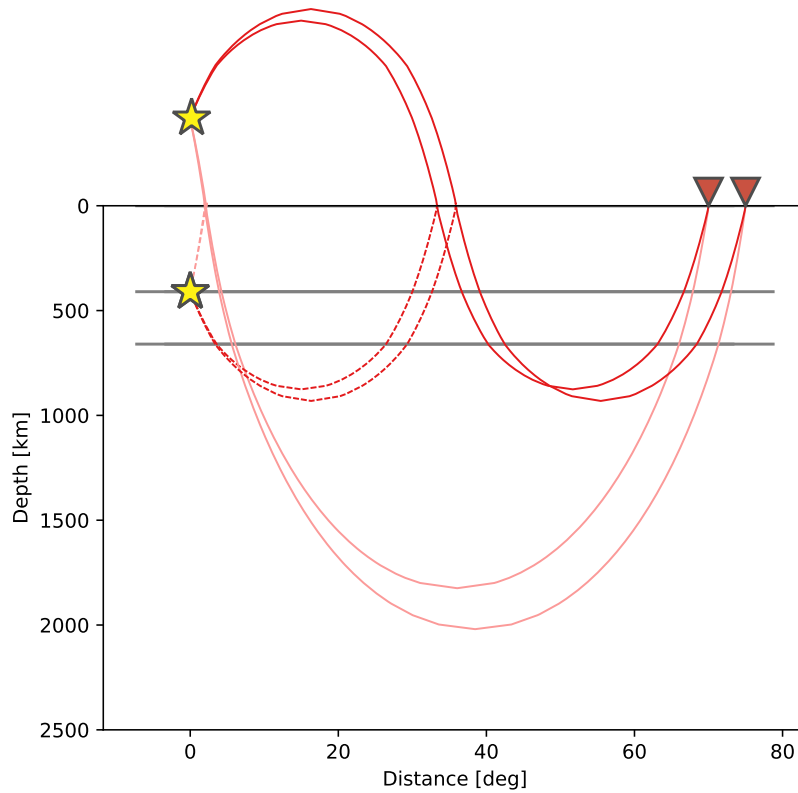


Figure 7.7: Schematic diagram of PP waves passing through the caustic. In this figure, we can see that the PP waves intersect, while the pP waves do not.

§7.6 Travel time analysis

Although the generating parameter σ provides insight into this system, it is not directly related to observables. In this section, relations among observables: epicentral distance $X(p_x)$, travel time $T(p_x)$, and ray parameter p_x are presented for a stratified medium^{note 11)}.

At the origin $\mathbf{p} = (p_x, \sqrt{\alpha(0)^{-2} - p_x^2})$. The deepest point of the ray z is given by $z = Z(p_x)$. In this case, the corresponding seismic velocity is represented by

$$\alpha(Z) = p_x. \quad (7.46)$$

z component of equation 7.13 leads to the following relation:

$$\frac{dz}{d\sigma} = p_z. \quad (7.47)$$

Let σ be the Σ it takes to reach the deepest point. Since the distance to the deepest point is half of $X(p_x)$, the x component of equation 7.13 becomes

$$\frac{X(p_x)}{2} = \int_0^\Sigma p_x d\sigma = \int_0^{Z(p_x)} \frac{dz}{p_z}. \quad (7.48)$$

^{note 12)} To summarise this,

$$X(p_x) = 2p_x \int_0^{Z(p_x)} \frac{1}{(\alpha^{-2}(z) - p_x^2)^{1/2}} dz \quad (7.49)$$

Next, consider the travel time $T(p_x)$. Since we are now thinking in σ , let dt relate to generating parameter with the equation 7.11,

$$dt = \frac{ds}{\alpha} = \frac{d\sigma}{\alpha^2}. \quad (7.50)$$

As in the case of X , by transforming $d\sigma$ into the integral of dz as well as X , T can be represented by

$$T(p_x) = \int_0^{s_d} \frac{ds}{\alpha} = 2 \int_0^{Z(p_x)} \frac{\alpha^{-2}}{(\alpha^{-2}(z) - p_x^2)^{1/2}} dz, \quad (7.51)$$

where s_d is the distance of the ray path from the source to the deepest point.

Finally, let us evaluate τ . Recall the definition:

$$\tau(p_x) = T(p_x) - p_x X(p_x). \quad (7.52)$$

^{note 11)} For details, read the textbook by Shearer⁽⁹⁾

^{note 12)} Since p_x is a conserved quantity, it is often confusing to write $T(p_x) = \int p_x dX = p_x X(p_x)$. In this case, $T(p_x) = \int \mathbf{p} \cdot d\mathbf{s}$ must be evaluated. This is an example of how inconsistency can arise when considering the physical image of a moving particle if we do not take the generating parameter σ as the parameter instead of the time T .

Substituting the equation derived above, we obtain

$$\tau(p_x) = 2 \int_0^{Z(p_x)} \sqrt{\alpha(z)^{-2} - p_x^2} dz, \quad (7.53)$$

τ has the advantage that the divergence to infinite disappears from the integral, making the actual evaluation easier.

For typical structures below, let us review the relationship between ray paths and travel time.
note 13)

7.6.1 In a case of monotonically increasing seismic velocity with depth

In the simplest case, let us consider travel time for monotonically increasing seismic velocity with depth. Figure 7.8 shows such an example. The ray parameter p_x conserves along the ray path. For this reason, the seismic velocity at the turning depth matches the corresponding apparent horizontal velocity ($1/p_x$). The ray path dives to the turning depth, then it returns to the surface.

This figure shows that with decreasing parameter $p_x = \sin \theta / \alpha(z)$ (decreasing the emergency angle equivalently), the ray reaches farther (the epicentral distance $X(p)$ becomes longer). In this case, $dX/dp_x < 0$, as shown in the figure, is referred to as prograde. With increasing epicentral distance, the ray density decreases. (This means that ray density can be estimated by dp_x/dX as shown in this figure.) Therefore, the amplitude decreases with the epicentral distance.

note 13) If you are interested in travel time calculations for real one-dimensional structures, please refer to [(3), (4), Crotwell et al. 1999, Buland and Chapman, 1983]

Problem 7.2

1. Let us consider wave propagations in upper xz plane ($z > 0$). When the seismic velocity α is a linear function given by $a + bz$, show that the ray path becomes a circle given by

$$\left(x - \frac{\sqrt{1 - p_x^2 a^2}}{b p_x}\right)^2 + \left(z + \frac{a}{b}\right)^2 = \frac{1}{p_x^2 b^2}. \quad (7.54)$$

Hint: The equation can be derived from the Hamilton equations and the Eikonal equation.

2. Derive equation 7.51.
3. Show that $T(p_x)$ is given by

$$T(p_x) = \frac{2}{b} \cosh^{-1} \left(\frac{1}{p_x a} \right). \quad (7.55)$$

4. We can calculate a ray path by numerically integrating the Hamilton equation with respect to σ . For example, ray paths in Figure 7.8 are calculated from the integration. Calculate the ray paths numerically in the same manner. Compare the analytic solution of the ray path and the numerical calculation.

* **Note:** The first parameter of the Hamilton equation describes the change of the location, whereas the second one determines the slowness vector as conserved the total energy along the ray path (in other words, to satisfy the Eikonal equation).

Hint:

$$\frac{d \cosh^{-1}(z)}{dz} = \frac{1}{\sqrt{z^2 - 1}} \quad (7.56)$$

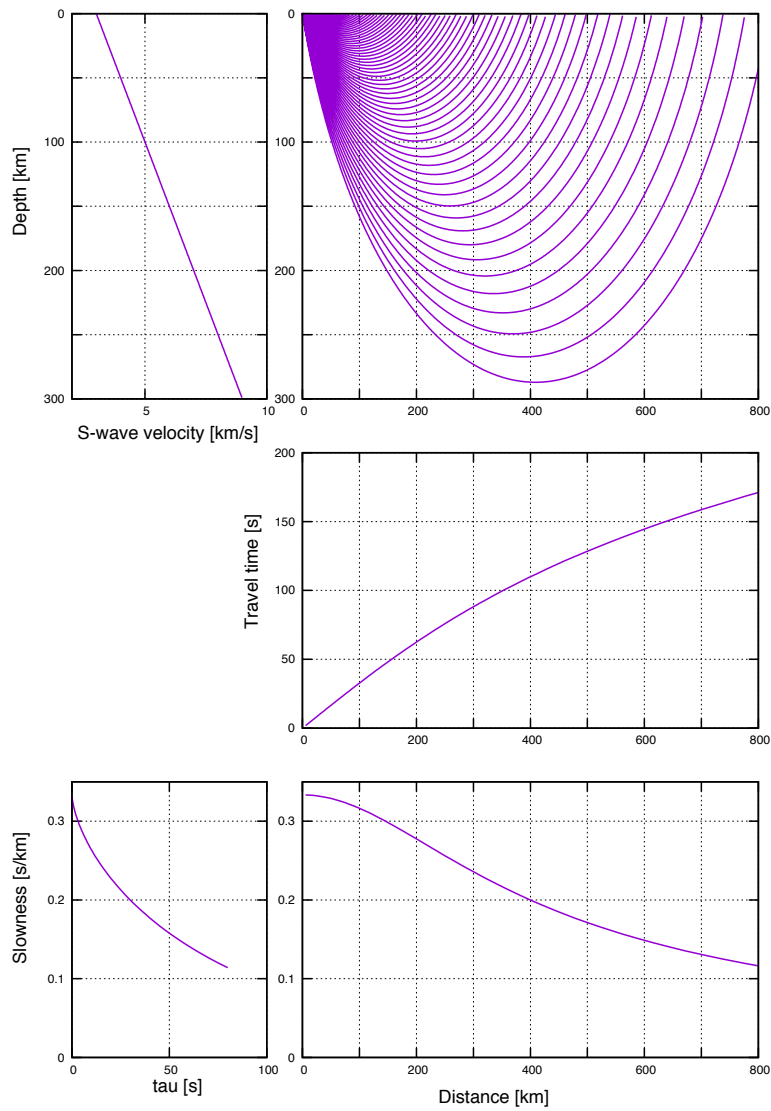


Figure 7.8: Ray paths for linearly increasing seismic structure with depth. We can see a relation of $p_x = dT/dx$.

7.6.2 In a case of a positive seismic jump at a depth

Next, let us consider a positive jump of seismic velocity at a depth.

When a ray path exists in the upper layer, it reaches further with the decreasing emergency angle (prograde $dX/dp < 0$). When the ray enters the layer of steeply increasing seismic velocity, the ray path backs to the source side according to Snell's law. This feature characterized by $dX/dp > 0$ is referred to as a retrograde ray. When the emergency angle decreases furthermore, it dives into the deeper lower layer, and the ray path becomes prograde again. At the point where $dX/dp = 0$, known as a caustic, the ray path density diverges, causing the amplitudes to diverge. Due to this singularity, ray theory cannot accurately handle the wavefield around the point where $dX/dp = 0$ ^{note 14}.

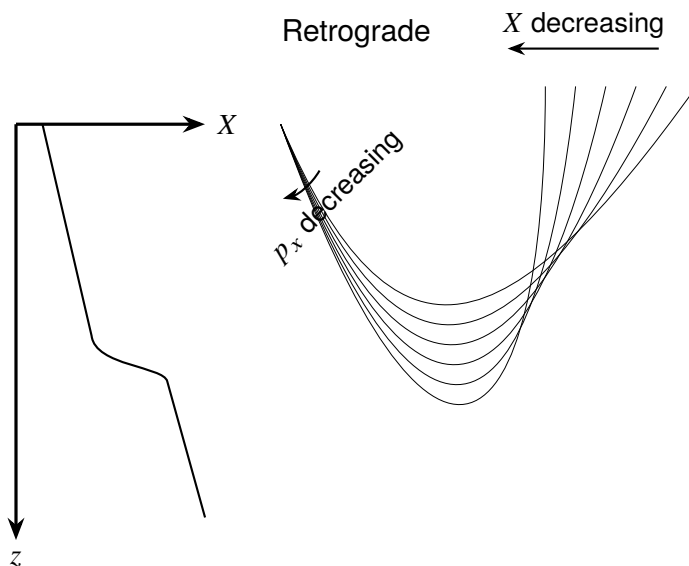


Figure 7.9: Ray paths involving rapidly increasing velocity zone.

Basic features, in this case, are common with those in a two-layer medium discussed in the previous chapter.

^{note 14} Because the high-frequency approximation is broken down at the point close to the caustic, we must consider an appropriate solution of the wave equation which connects to the ray theoretical solution. Such a solution leads to jumping off the phase of 90° after the passage through the caustic. The phase jump is also discussed in the later section about amplitudes of a ray theoretical solution.

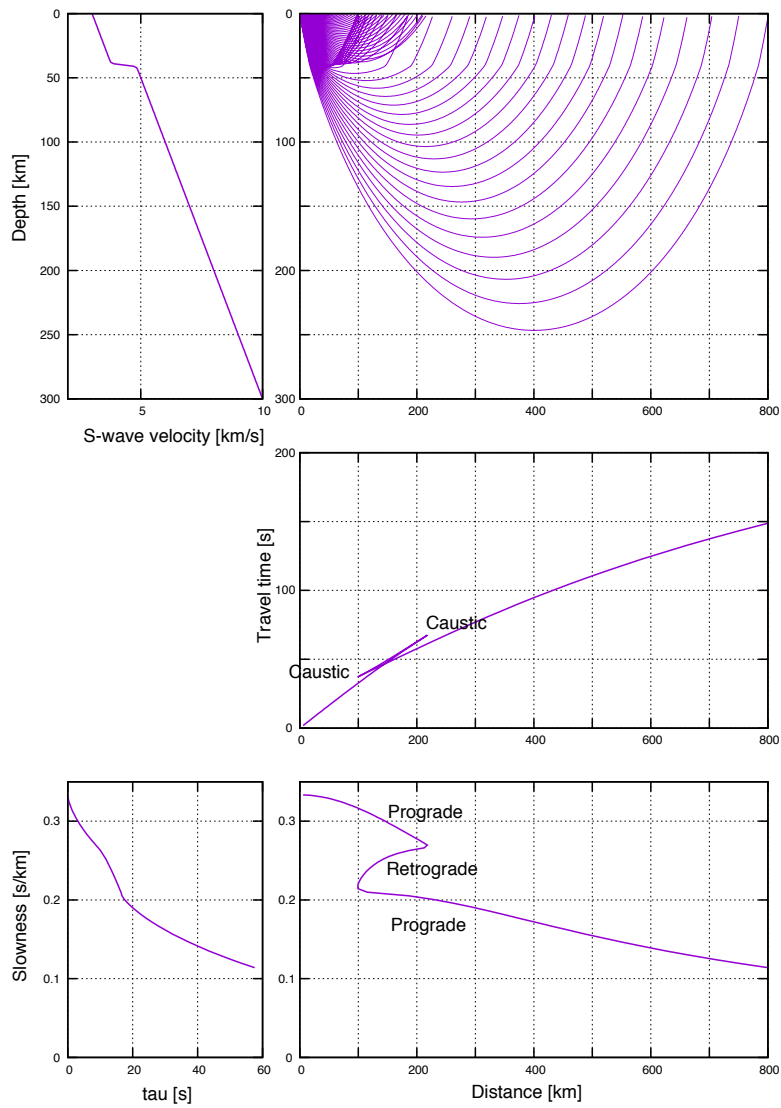


Figure 7.10: Ray paths in a case of positive seismic jump at a depth.

If you only look at the wavefront, you may find it difficult to see the triplication area because they overlap each other. Therefore, let's zoom in on the ray near the triplication area with the wavefront at the same time (Figure 7.11). The retrograde ray corresponds roughly to the green color. The caustic surface is formed at the point where the retrograde starts. When triplication occurs, the rays are folded. The wedge-shaped base of the folded wavefront corresponds to the wavefront that passes through the caustic surface, resulting in a 90-degree phase shift. You can also see that the positive velocity gradient just below the discontinuity bends the rays toward the surface, which increase the amplitude of the retrograde branch. Thus, when considering seismic waves, it is important to consider both the rays and the wavefront together. You can also see that in the $\tau - p$ region, triplication is unfolded and becomes a single-valued function.

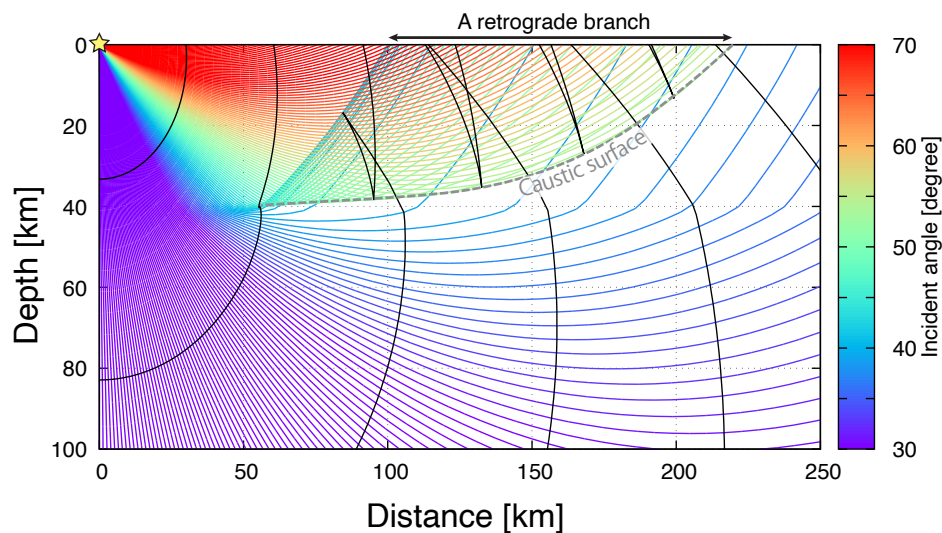


Figure 7.11: Wavefront with positive seismic velocity jump. The emergence angle is shown in color, and the wavefront (isochrones of travel time) is also shown

Consider another example. It is known that a 410 km discontinuity and a 660 km discontinuity exist globally in the Earth's interior. These two discontinuities correspond to phase changes in minerals (corresponding to olivine $\rightarrow$ spinel and spinel $\rightarrow$ perovskite, respectively). Let's take a look at Figure 7.12. Here is a seismic waveform record in Alaska of an earthquake that occurred in the Aleutians at a depth of about 100 km. The waveforms are complicated, but we can see multiple P-wave packets arriving in $13^\circ \sim 30^\circ$. This is the result of triplication at the 410 km and 660 km discontinuities. The lower figure shows the theoretical travel time. You can see that it roughly represents the characteristics of the observed waveforms.

Let us imagine that there is no information on the Earth's interior. From Figure 7.12, we can see that it is difficult to interpret the waveforms of the P wave except at the first arrival time when there are multiple arrivals. In the figure, this corresponds to the blue rising edge. The rise of the first arriving wave packet is easy to read (because the ground is quiet before the arrival) and can be measured with good accuracy. If we only have information on the first arrivals, it is difficult to

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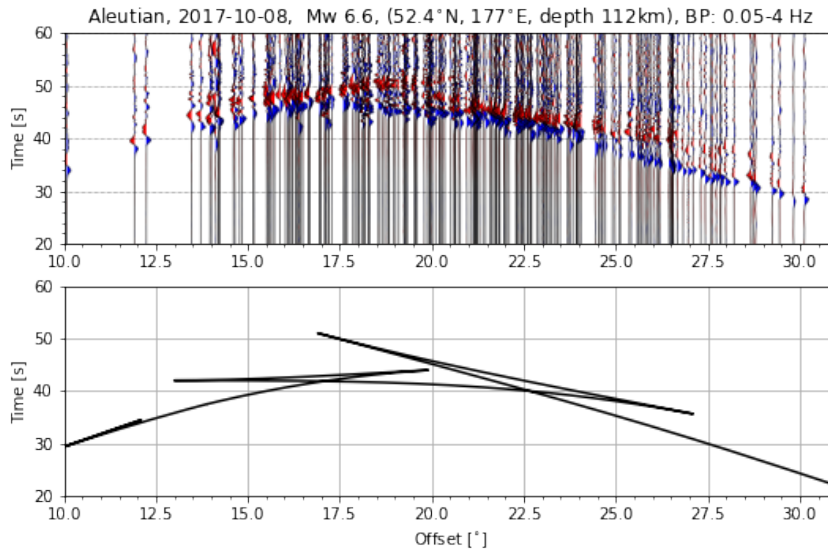


Figure 7.12: A typical example with a positive seismic velocity jump. Velocity waveform (0.05-4 Hz) observed by USArray (Alaska) for an earthquake that occurred in the Aleutians. For a better display, the travel time of the waveform was reduced (specifically, $T(10 \text{ km/s})X$ is plotted against X). Positive amplitudes are shown in red and negative amplitudes are shown in blue. It can be seen that there are multiple arrivals of P-wave wave packets in this distance range. The lower figure also shows the corresponding theoretical travel times, where we can see the triplication corresponding to the positive jumps in the two seismic velocity jumps (410 km discontinuity and 660 km discontinuity).

distinguish it from the travel time curve for a structure without the discontinuities (e.g, 7.8). In other words, when inferring the internal structure of the earth from the travel time, without information on triplication, the information on the internal discontinuity cannot be correctly estimated, and an over smooth structure will be inferred (since the simple model can also explain the measured first arrivals).

Problem 7.3

Let us consider wave propagation in xz plane ($z > 0$) for seismic speed α given by

$$\alpha(z) = 5 + (1 + \tanh(z - 40)) * 2 + \frac{z}{200}. \quad (7.57)$$

By numerical integration of Hamilton equations with respect to σ , calculate the ray paths. And as in Figure 7.9, plot the relations between Travel time and distance, that between distance and slowness, and that between τ and slowness. Then compare the results with two-layered mediums.

7.6.3 In a case of a negative seismic jump at a depth

When a negative jump of seismic velocity with depth exists, rays cannot reach a certain area on the surface known as a shadow zone. Because rays tend to avoid the low-velocity zone, as shown in Figure 7.13, the seismic exploration of the seismic structure becomes difficult in general.

§7.7 1-D inversion

7.7.1 Herglotz-Wiechert inversion

If the seismic velocity structure monotonically increases with depth, we can infer the seismic velocity structure from travel time information. For a given slowness p_x , the corresponding arrival distance $X(p_x)$ is

$$X(p_x) = 2p_x \int_0^{z(p_x)} \frac{dz}{\sqrt{\alpha^{-2}(z) - p_x^2}}. \quad (7.58)$$

Although it may seem abrupt, when the seismic velocity increases monotonically, we can simply transform the integration variable from z to $\eta = 1/\alpha$, which gives

$$X(p_x) = 2p_x \int_{\eta_0}^{p_x} \frac{1}{\sqrt{\eta^2 - p_x^2}} \frac{dz}{d\eta} d\eta. \quad (7.59)$$

Suppose the slowness of the seismic wave observed at a certain distance X_1 is p_1 . Dividing both sides by $\sqrt{p_x^2 - p_1^2}$, and integrating with respect to p_x from p_1 to η_0 yields (omitting the details⁽⁸⁾) the depth z_1 where the seismic velocity is $1/p_1$:

$$z_1 = \frac{1}{\pi} \int_0^{X_1} \cosh^{-1}(p_x(X)/p_1) dX. \quad (7.60)$$

This relation is known as the Herglotz-Wiechert formulas. It is important to note that when the travel time T is observed as a monotonically increasing function of distance X , the seismic velocity structure can be uniquely represented by an integral of the observational data. While highly elegant, its application conditions are strict, leaving only limited examples where it can be applied directly.

7.7.2 τ -p inversion

Let us now consider the problem of inferring the structure of the Earth's interior from the travel time of seismic waves measured at the Earth's surface. Suppose we now have a perfectly measured travel time $T(X)$. In this case, the slowness p at location X is

$$p = \frac{dT}{dX}, \quad (7.61)$$

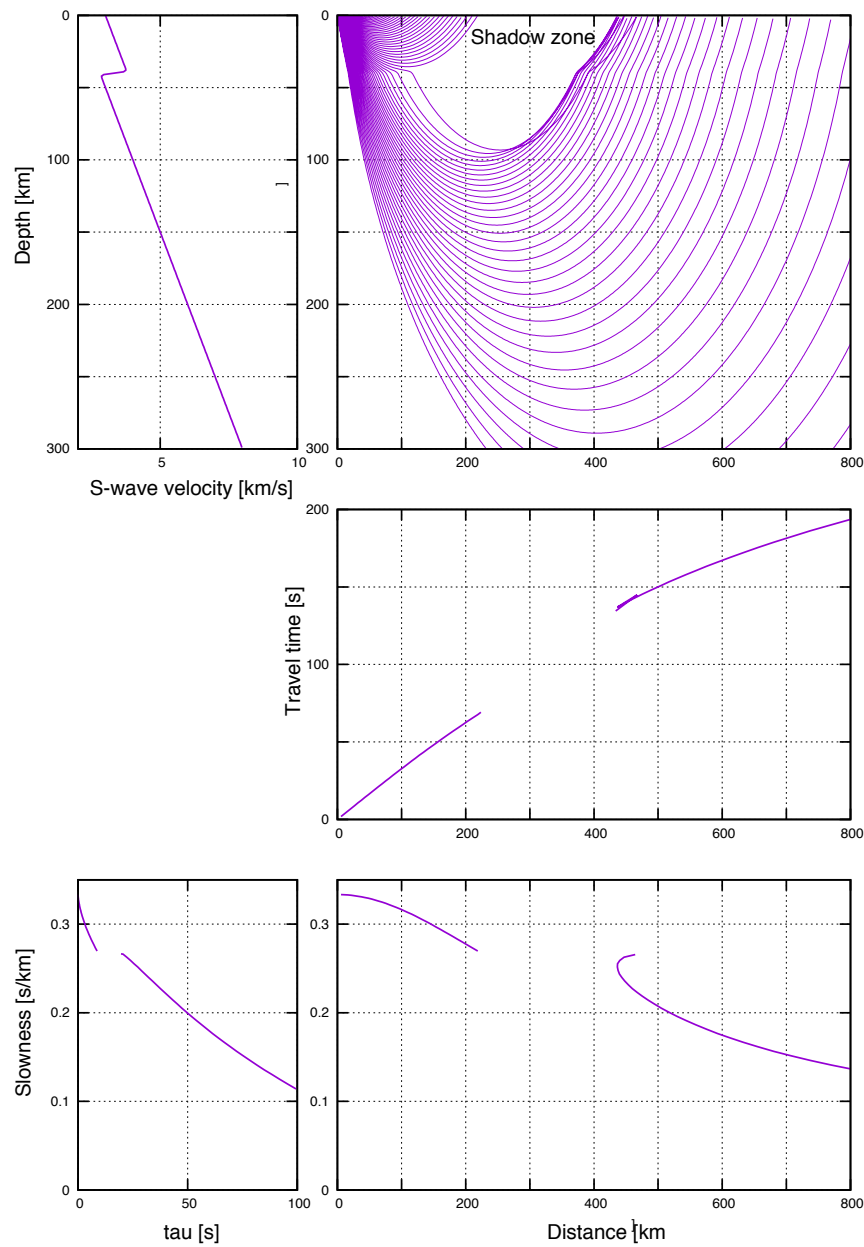


Figure 7.13: Ray paths in a case of negative seismic jump at a depth.

and can be obtained from its direction. The τ can be measured as the interception time (y-intercept) at which the tangent line is extended and intersects at $X = 0$.

Let us assume that the Earth's internal structure consists of n homogeneous layers. Let the <https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

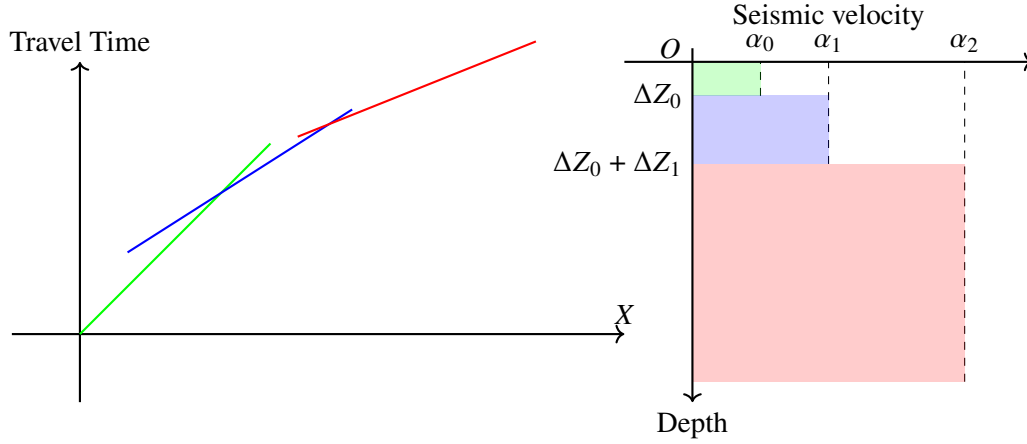


Figure 7.14: Schematic figure of a segmented line approximation of the travel time curve.

thickness of the n layer be ΔZ_n and the seismic wave velocity be α_n (equation 7.14). Discretising the equation for τ (equation 7.53), we obtain

$$\tau(p_x) = 2 \sum_{i=1}^n \sqrt{\alpha_n^{-2} - p_x^2} \Delta Z_n. \quad (7.62)$$

We can find here that if α_n is known, it is a linear equation. In fact, α_n can be measured from the graph. If each line is a direct wave (red), a head wave in the second layer, and a head wave in the third layer, each slowness corresponds to α_n^{-1} . The we can obtain the following relation:

$$\begin{pmatrix} \tau_1 \\ \tau_2 \\ \vdots \\ \tau_n \end{pmatrix} = 2 \begin{pmatrix} \sqrt{\alpha_0^{-2} - \alpha_1^{-2}} & 0 & 0 & \cdots \\ \sqrt{\alpha_0^{-2} - \alpha_2^{-2}} & \sqrt{\alpha_1^{-2} - \alpha_2^{-2}} & 0 & \cdots \\ \cdots & \vdots & \cdots & \cdots \\ \sqrt{\alpha_0^{-2} - \alpha_n^{-2}} & \sqrt{\alpha_1^{-2} - \alpha_n^{-2}} & \cdots & \sqrt{\alpha_{n-1}^{-2} - \alpha_n^{-2}} \end{pmatrix} \begin{pmatrix} \Delta Z_0 \\ \Delta Z_1 \\ \vdots \\ \Delta Z_n \end{pmatrix}. \quad (7.63)$$

The linear form of the equation means that it is possible to solve the equation from the top layer.

Problem 7.4

Suppose that the travel time T [s] at a point X [km] can be measured at three locations. An earthquake occurred at $X = 0$ on the surface at time $T = 0$. Assume that each observation is $(X_0, T_0) = (15, 3.00)$, $(X_1, T_1) = (80, 13.40)$ and $(X_2, T_2) = (120, 18.20)$, approximate the travel time T by a line connecting these points. Estimate the Earth's internal structure by calculating the $\tau - p$ plot.

§7.8 Seismic tomography

Travel time information can be used to infer not only 1-D structure but also 3-D structure. This technique is called seismic tomography^{note 15)}. Let us consider a simplified case here. For simplicity, we assume a 2-D homogeneous medium (Figure 7.15). In the figure, triangles represent observation stations and a star represents a hypocenter. Since a homogeneous medium is assumed, rays are straight lines. Now consider the i -th ray path. Let the travel time anomaly corresponding to the i -th ray path be ΔT_i . As shown in the figure, suppose the region is divided into grids, and each grid position is specified by an index j . Suppose the i -th ray crosses the grid j with a length of l_{ij} . Considering the slowness anomaly Δp_j for each grid, the travel time anomaly ΔT_i can be written as

$$\Delta T_i = \sum_j l_{ij} \Delta p_j. \quad (7.64)$$

By solving this equation using the least squares method, we can determine the slowness anomalies Δp_j .

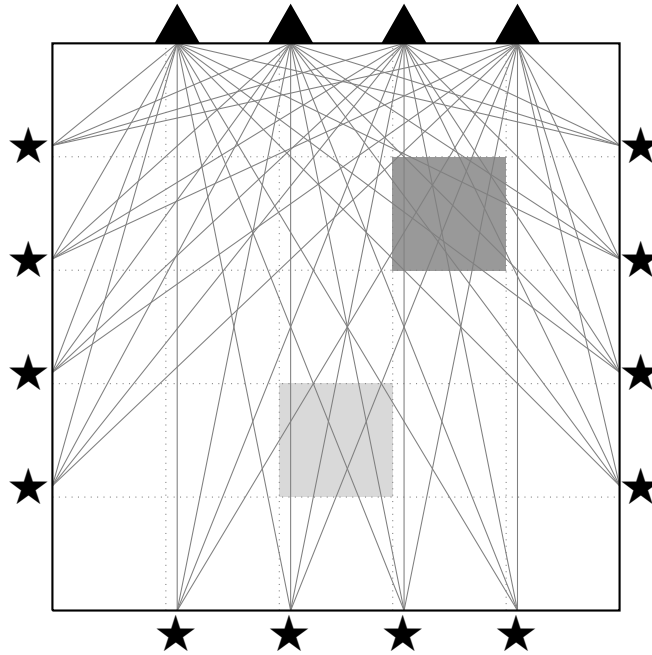


Figure 7.15: Schematic diagram of simplified seismic tomography. Triangles represent observation stations, and stars represent hypocenters. Rays are assumed to be straight lines.

^{note 15)}For details, refer to textbooks by Shearer⁽⁹⁾ and Nolet⁽⁶⁾

§7.9 Applicability of ray theory

Let us consider the applicability range of ray theory. In order for ray theory to be applicable, high frequency approximation must be valid. In other words, the seismic velocity structure must be sufficiently smooth. Let k be the wavenumber and a be the spatial scale of the velocity anomaly. The condition $ka \gg 1$ must be satisfied. However, considering actual velocity anomalies, it is known that velocity anomalies of various spatial scales exist. This implies that when a seismic wave propagates over a certain distance L , it will cross velocity anomalies of a scale comparable to its wavelength and undergo scattering. This distance is known as the mean free path L . When the propagation distance becomes significantly larger than L , multiple scattering occurs. This is like looking into the interior through frosted glass, making it practically difficult to consider rays. This is because the wave propagation is randomized due to scattering. Figure 7.16 shows a diagram known as the $ka - kL$ diagram. It is important to understand which approximations are reasonable and their application limits depending on each problem setting.

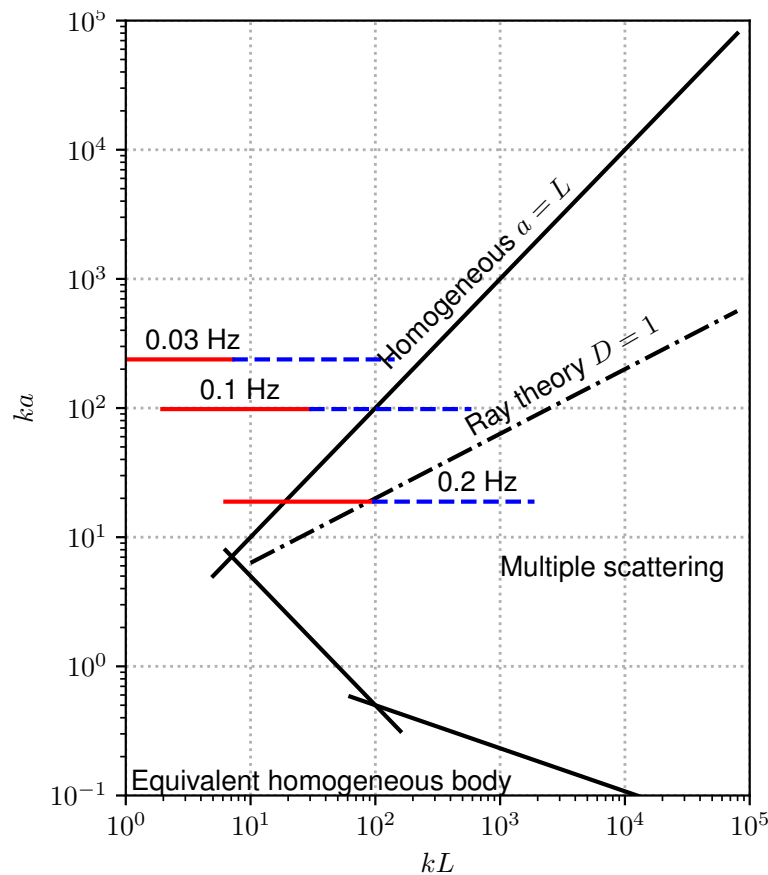


Figure 7.16: $ka - kL$ diagram.

§7.10 Tools for travel time analysis

When we calculate the travel time for a stratified Earth, `taup` (<http://www.seis.sc.edu/taup/>) is a common toolkit among seismologists. A direct Eikonal solver using the fast marching algorithm is also common. This algorithm is applicable to a complex medium (non-horizontally-stratified structure). Some different programs using fast marching algorithms are available at a website by Nick Rawlinson (<http://rses.anu.edu.au/~nick/>). `PyKonal`,⁽¹¹⁾ a Python package, is also available.

§7.A IASPEI standard phase list

For details, see <http://www.isc.ac.uk/standards/phases/>.

7.A.1 CRUSTAL PHASES

Pg	At short distances, either an upgoing P wave from a source in the upper crust or a P wave bottoming in the upper crust. At larger distances also, arrivals are caused by multiple P-wave reverberations inside the whole crust with a group velocity of around 5.8 km/s.
Pb	(alt:P*) Either an upgoing P wave from a source in the lower crust or a P wave bottoming in, the lower crust
Pn	Any P wave bottoming in the uppermost mantle or an upgoing P wave from a source in the uppermost mantle
PnPn	Pn free-surface reflection
PgPg	Pg free-surface reflection
PmP	P reflection from the outer side of the Moho
PmPN	PmP multiple free surface reflection; N is a positive integer. For example, PmP2 is PmPPmP
PmS	P to S reflection/conversion from the outer side of the Moho
Sg	At short distances, either an upgoing S wave from a source in the upper crust or an S wave bottoming in the upper crust. At larger distances also arrivals caused by the superposition of multiple S-wave reverberations and SV to P and/or P to SV conversions inside the whole crust.
Sb	(alt:S*) Either an upgoing S wave from a source in the lower crust or an S wave bottoming in, the lower crust
Sn	Any S wave bottoming in the uppermost mantle or an upgoing S wave from a source in the uppermost mantle
SnSn	Sn free-surface reflection
SgSg	Sg free-surface reflection
SmS	S reflection from the outer side of the Moho
SmSN	SmS multiple free-surface reflections; N is a positive integer. For example, SmS2 is SmSSmS
SmP	S to P reflection/conversion from the outer side of the Moho
Lg	A wave group observed at larger regional distances and caused by the superposition of multiple S-wave reverberations and SV to P and/or P to SV conversions inside the whole crust. The maximum energy travels with a group velocity of approximately 3.5 km/s
Rg	Short-period crustal Rayleigh wave

7.A.2 MANTLE PHASES

§7.B Stratified Earth models

You can find a reference stratified Earth model.

7.B.1 PREM

Preliminary Reference Earth Model <http://ds.iris.edu/ds/products/emc-prem/>

7.B.2 AK135

<http://ds.iris.edu/ds/products/emc-ak135-f/>
<http://rses.anu.edu.au/seismology/ak135/intro.html>

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P	A longitudinal wave, bottoming below the uppermost mantle; also an upgoing longitudinal wave from a source below the uppermost mantle
PP	Free-surface reflection of P wave leaving a source downward
PS	P, leaving a source downward, reflected as an S at the free surface. At shorter distances, the first leg is represented by a crustal P wave.
PPP	analogous to PP
PPS	PP, which is converted to S at the second reflection point on the free surface; travel time matches that of PSP
PSS	PS reflected at the free surface
PcP	P reflection from the core-mantle boundary (CMB)
PcS	P converted to S when reflected from the CMB
PcPN	PcP are reflected from the free surface N-1 times; N is a positive integer. For example PcP2 is PcPPcP
Pz+P	(alt:PzP) P reflection from the outer side of a discontinuity at depth z; z may be a positive numerical value in km. For example, P660+P is a P reflection from the top of the 660 km discontinuity.
Pz-P	P reflection from the inner side of a discontinuity at depth z. For example, P660-P is a P reflection from below the 660 km discontinuity, which means it is precursory to PP.
Pz+S	(alt:PzS) P converted to S when reflected from the outer side of discontinuity at depth z
Pz-S	P converted to S when reflected from the inner side of discontinuity at depth z
PScS	P (leaving a source downward) to ScS reflection at the free surface
Pdif	(old:Pdiff) P diffracted along the CMB in the mantle
S	Shear wave, bottoming below the uppermost mantle; also an upgoing shear wave from a source below the uppermost mantle
SS	Free surface-reflection of an S wave leaving a source downward
SP	S, leaving a source downward, reflected as P at the free surface. At shorter distances, the second leg is represented by a crustal P wave.
SSS	analogous to SS
SSP	SS converted to P when reflected from the free surface; travel time matches that of SPS.
SPP	SP reflected at the free surface
ScS	S reflection from the CMB
ScP	S converted to P when reflected from the CMB
ScSN	ScS multiple free-surface reflections; N is a positive integer. For example ScS2 is ScSScS
Sz+S	(alt:SzS) S reflection from the outer side of a discontinuity at depth z; z may be a positive numerical value in km. For example, S660+S is an S reflection from the top of the 660 km discontinuity.
Sz-S	S reflection from the inner side of discontinuity at depth z. For example, S660-S is an S reflection from below the 660 km discontinuity, which means it is precursory to SS.
Sz+P	(alt:SzP) S converted to P when reflected from the outer side of discontinuity at depth z
Sz-P	S converted to P when reflected from the inner side of discontinuity at depth z
ScSP	ScS to P reflection at the free surface
Sdif	(old:Sdiff) S diffracted along the CMB in the mantle